Strengthening the Low Frames of a Nine-Point Certificate for Simple Critical Zeros of the Riemann Zeta Function

Li Xiang
lixiang90

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Abstract

The outer case of a nine-point certificate can be improved without enlarging its strong low-frame capture. We strengthen ten of the 241 captured eight-point frames, using 35 additional occurrences of 32 points in the original certified kernel table. The other 231 frames retain their original exact duals. This proves a larger global eight-point floor, which raises the high–low average, while the previous complete low–low certificate already pays the new target. The resulting uniform nine-point reward is $8054119/10^9$. The unchanged lossless all-zero transport gives

$$\liminf_{T\to\infty}\frac{N_0^s(T)}{N(T)}\geq\frac{668124910}{991945881}=0.673549759918807505\ldots.$$

The numerator counts simple critical-line zeros; the denominator counts all nontrivial zeros with multiplicity. The new finite implication is checked with exact rational arithmetic under the retained continuous point, convexity and capture certificate. The preceding bound has a complete ordinary Lean proof and independent kernel replay. The additional low-frame strengthening has not yet been ported to Lean or accepted by the record website.

1 Fixed framework and statement

Let $N(T)$ count all nontrivial zeros of ζ with imaginary part in $(0,T]$, including multiplicity, and let $N_0^s(T)$ count the simple zeros on $\Re s=1/2$ in that interval.

Theorem 1.1. *Within the retained continuous AM certificate and proved lossless all-zero analytic transport,*

$$\liminf_{T\to\infty}\frac{N_0^s(T)}{N(T)}\geq\frac{668124910}{991945881}.$$

The new input is the exact finite low-frame implication in Section 3. No additional correlation conjecture, zero-free region or arithmetic moment assumption is used.

The preceding bound was $941021/1397107=66812491/99194597$ [2]. The present gain is exactly

$$\frac{668124910}{991945881}-\frac{941021}{1397107}=\frac{83750869}{1385854533966267},$$

or approximately 0.00000604326550495 percentage points. This is a comparison with the project's fixed preceding result. No exhaustive global priority or official record acceptance is asserted.

Put $\beta=1/\sqrt{2}$, $Z=\sqrt{2}\sin\beta$ and $I=[-1/2,1/2]$. With zero extension outside I, the unchanged positive normalized density and its real even kernel are

$$f(u)=\frac{\cos(\sqrt{2}u)+\sum_{j=1}^{12}c_j\cos(2\pi ju)}{Z},\qquad K(t)=\int_I f(u)e^{-2\pi itu}\,du,\qquad w(t)=K(t)^2,$$

where the integer numerators $10^9 c_j$ are

$$(12310798, -15041681, 3867664, 6489926, -992327, 5580097,$$
$$-6846472, 3781297, -5670353, 3355089, -450523, -218483).$$

In particular $0 \le w(t) \le 1$ on $\mathbb{R}$. For seven real gaps $h_r \ge \theta = 4/5$, define

$$F_8(h) = \sum_{r=0}^{6} b_r h_r + \sum_{0 \le i < j \le 7} a_{ij} w(h_i + \cdots + h_{j-1}),$$

where $10^8 b_r = (28898, 57272, 75526, 80958, 75526, 57272, 28898)$. The successive rows of $10^8 a_{ij}$ are

$$(13630830, 0, 19565904, 20317441, 45030671, 100000000, 200000000),$$
$$(29997996, 30621878, 47766489, 79682558, 109938656, 100000000),$$
$$(37356140, 69378121, 65335210, 79682558, 45030671),$$
$$(38030065, 69378121, 47766489, 20317441),$$
$$(37356140, 30621878, 19565904), \quad (29997996, 0), \quad (13630830).$$

There are 26 nonzero pair coefficients. Both the gap coefficients and the pair coefficients are invariant under reflection. For eight gaps $g_r \ge \theta$, put

$$F_9(g) = \tfrac{1}{2}\{F_8(g_0, \ldots, g_6) + F_8(g_1, \ldots, g_7)\} + 2w\left(\sum_{r=0}^{7} g_r\right).$$

2 The retained cover and the new outer case

We retain the following proved continuous inputs [1, 2, 3]. First, any frame with $F_8 < \tau$, where $\tau = 805803/10^8$, belongs to one of 241 closed captured cells or its reflection. Second, for all compatible pairs of captured cells, the full real nine-point functional has the uniform lower bound

$$M = \frac{263917374155379049835090394947}{32768000000000000000000000000000}.$$

The latter certificate has 2,399 physical pair domains and 84 closed branches replacing the 47 weakest domains. It includes all finite-box dual residuals. Its precise continuous implication is retained; the stored minimum is rechecked from the existing rational witnesses.

Lemma 2.1. *On every captured cell and every reflected captured cell,*

$$F_8 \ge \ell := 805021/10^8.$$

Consequently this inequality holds for every real seven-gap frame with gaps at least $4/5$.

Proof. The finite assertion is proved in Section 3. Reflection preserves the complete functional, as well as the reflected cell constraints. A frame outside the captured union has $F_8 \ge \tau$ by the contrapositive of complete capture, and $\tau > \ell$. The cover is continuous: integer endpoints encode rational inequalities; the real gaps are not restricted to a lattice. □

Proposition 2.2. *For every real eight-gap frame with* $g_r \ge 4/5$,

$$F_9(g) \ge c := 8054119/10^9.$$

Proof. If both eight-point subframes belong to the captured union, their compatible pair is covered by the retained low–low certificate, and $F_9 \ge M > c$. Otherwise one subframe has $F_8 \ge \tau$. Lemma 2.1 bounds the other by ℓ; the extra total-span square is nonnegative. Hence

$$F_9 \ge (\tau + \ell)/2 = 805412/10^8 > c.$$

This argument includes frames with equality at the capture threshold and the shared endpoints of closed cells. It strengthens the actual global eight-point floor, rather than extrapolating a finite low–low margin past an unchanged outer-case bound. □

3 Exact low-frame certificate

For each original captured cell we reconstruct the original YA lower bound without rounding. Only the following ten zero-based labels have this original lower bound below ℓ:

$$17, 53, 64, 74, 111, 131, 148, 166, 183, 213.$$

The other 231 original cells already pay ℓ and retain their original witnesses. The ten replacement problems have 33 variables: seven real gaps and the 26 physical squares attached to nonzero pair coefficients. Original span inequalities, gap boxes and all original valid affine minorants are retained. Each physical square has its complete interval $[0, 1]$.

The additional affine rows use 35 occurrences of 32 original PTL points, with scale $SC = 32768$. The complete catalog records the original point numerators, packs and PTL list indices. Four numerators, $98272, 230769, 231200, 265088$, are absent from the preceding formal development's 1,089-point catalog; all 32 are entries of the original 2,196-point table. No new analytic kernel data are generated. The source identity, point packs and convexity-region packs are checked against the fixed original source [1].

For completeness, if a proved entry gives $v \le w(p)$ and $d_- \le w'(p) \le d_+$, and the certified convexity interval contains $[l_*, u_*]$ where $l_* := \min(l, p)$ and $u_* := \max(u, p)$, then for all $x \in [l, u]$ the two valid affine bounds are

$$w(x) \ge v + d_+(l_* - p) + d_-(x - l_*),$$
$$w(x) \ge v + d_-(u_* - p) + d_+(x - u_*).$$

They follow from the genuine convex tangent and the signs of the two endpoint corrections. The checker includes both the query interval and the point in the certified convexity interval. A point outside the smaller query interval is used only with these valid reanchoring corrections.

Write the reconstructed finite constraints as $Ax \le b$, $L \le x \le U$, and the numerator of $10^8 F_8$ as $q^T x$. For each exact nonnegative rational multiplier vector λ, set $r = q + A^T\lambda$. Then

$$F_8 \ge \frac{-\lambda^T b + \sum_i r_i(L_i \text{ if } r_i \ge 0;\ U_i \text{ otherwise})}{10^8}.$$

Every nonzero residual is retained, regardless of size. Across all ten patches the positive multiplier support has 287 entries. Some patches need only a better dual of the original rows; the total number of added tangent rows is 35.

The smallest strengthened bound is checked exactly from all ten patches and 231 retained original witnesses, and is at least ℓ. Hence $(\tau + \ell)/2 > M > c$, where

$$M - c = \frac{2763379049835090394947}{32768000000000000000000000000000} > 0.$$

The same finite target is therefore paid by both parts of the cover. A floating linear program proposes multipliers only; the accepted implication uses the exact rational rows, signs and complete boxes.

4 All-zero counting and verification scope

The unchanged sum of gap pressures is $B = 404350/10^8$. Sliding the nine-point frame preserves every index-span pair budget at most two, the pair-mass bound below sixteen, and the endpoint cost $8c$. The original near-pair reward exceeds $1/40 > c$, and the original band-limited majorant has mass below two. The smoothing loss remains $32d\varepsilon$. All off-line reflected pairs and multiplicities are retained in the Hermitian Gram argument [2, 4]. First let $T \to \infty$ at fixed

smoothing, then let the smoothing parameter decrease to zero. The proved transport and the unchanged energy bound give

$$\liminf \frac{N_0^s(T)}{N(T)} \geq \frac{2 - C_{AM} - B}{1 - c} > \frac{668124910}{991945881}.$$

This proves Theorem 1.1. Simple-zero inclusion also gives the same weaker lower bound for distinct critical-line zeros.

The public checker and frozen data are stored in the mathematical repository. Its ordinary check requires only the Python standard library and the existing public rational helpers. With the original source available, an optional source check matches all 32 entries and their original list indices. The prior complete nine-point certificate is reused; no new search of its 2,399 domains is needed. Optimized Python is rejected because assertions are essential.

The complete ordinary Lean theorem and independent NaNoDa replay of the preceding 941021/1397107 bound are available at fixed commit `c705c4a` [3]. They retain the previous proofs, submissions and verification records. This new low-frame arithmetic implication has not yet been ported to that formalization. Exact rational replay, full formal proof, website resource compliance and official acceptance are separate verification claims. No new zero-free region is asserted here.

References

[1] *AM eight-point PC8CL certificate*, fixed public submission `dani-bound-2026`, commit `d272437e7ebeb17b87c8c3d7cece592aa23785ab`, original proof source.

[2] Li Xiang, *The Total-Span Term in a Nine-Point Certificate for Simple Critical Zeros of the Riemann Zeta Function*, October 10, 2026, fixed manuscript.

[3] Li Xiang, *Closed ninth-span formalization*, October 11, 2026, fixed formalization and verification records.

[4] Li Xiang, *Lossless Assembly of an Eight-Point Certificate for Simple Critical Zeros of the Riemann Zeta Function*, October 8, 2026, fixed manuscript.